

Chi squared goodness of fit tests



1 D, C, G, F, E, A, B or D, C, G, E, F, A, B

2

- a The null hypothesis is that the distribution of students has not changed.
 $H_0 : p_1 = 0.55, p_2 = 0.30, p_3 = 0.15, H_1 : H_0$ is rejected and at least one of the following is true: $p_1 \neq 0.55, p_2 \neq 0.30, p_3 \neq 0.15$

- b $\alpha = 0.05$

c

	No regular exercise	Occasional exercise	Regular exercise	Total
Observed number of students	113	72	25	210
Expected number of students	115.5	63	31.5	210

- d $df = 2$

- e $\chi^2 = 2.6811, p = 0.2617$

- f $\alpha = 5\%, p > \alpha$, therefore we do not reject the null hypothesis. It appears that there has been no change in exercise levels.

3

- a The null hypothesis is that the distribution of smartie colours is even.

$H_0 : p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = p_7 = p_8 = \frac{1}{8}, H_1 : H_0$ is rejected and at least one of the following is true: $p_1 \neq \frac{1}{8}, p_2 \neq \frac{1}{8}, p_3 \neq \frac{1}{8}, p_4 \neq \frac{1}{8}, p_5 \neq \frac{1}{8}, p_6 \neq \frac{1}{8}, p_7 \neq \frac{1}{8}, p_8 \neq \frac{1}{8}$

- b $\alpha = 0.05$

- c 678.5 smarties

- d $df = 7$

- e $\chi^2 = 72.86$

- f $p = 3.89 \times 10^{-13}$

- g $\alpha = 0.05, p < \alpha$, therefore we reject the null hypothesis. It appears that the distribution of smartie colours is not even.

4



- a The null hypothesis is that the die is fair.

$H_0 : p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = p_7 = p_8 = \frac{1}{8}$, $H_1 : H_0$ is rejected and at least one of the following is true: $p_1 \neq \frac{1}{8}, p_2 \neq \frac{1}{8}, p_3 \neq \frac{1}{8}, p_4 \neq \frac{1}{8}, p_5 \neq \frac{1}{8}, p_6 \neq \frac{1}{8}, p_7 \neq \frac{1}{8}, p_8 \neq \frac{1}{8}$

- b $\alpha = 0.02$

c

	1	2	3	4	5	6	7	8
Observed freq	8	11	12	10	13	11	15	8
Expected freq	11	11	11	11	11	11	11	11

- d $df = 7$

- e $\chi^2 = 3.64$

- f $p = 0.82$

- g $p > \alpha$ therefore we do not reject the null hypothesis. It appears that the die is fair.

5

- a The null hypothesis is that the city proportion is the same as the school proportion.

$H_0 : p_1 = 0.47, p_2 = 0.37, p_3 = 0.10, p_4 = 0.06$, $H_1 : H_0$ is rejected and at least one of the following is true: $p_1 \neq 0.47, p_2 \neq 0.37, p_3 \neq 0.10, p_4 \neq 0.06$

- b $\alpha = 0.10$

c

	O	A	B	AB
Observed frequency	88	57	21	6
Expected frequency	80.84	63.64	17.20	10.32

- d $df = 3$

- e $\chi^2 = 3.97$

- f $p = 0.26$

- g $p > \alpha$ therefore we do not reject the null hypothesis. It appears that the blood type proportions for the school do not differ significantly from that of the city.

6

- The null hypothesis is that the spinner is fair. $H_0 : p_1 = \frac{25}{36}, p_2 = \frac{1}{6}, p_3 = \frac{5}{36}$
The alternative is that H_0 is rejected and at least one of the following is true:
 $p_1 \neq \frac{25}{36}, p_2 \neq \frac{1}{6}, p_3 \neq \frac{5}{36}$. The significance level $\alpha = 0.05$.

	Green	Red	Blue
Observed frequency	142	31	27
Expected frequency	138.89	33.33	27.78

- $df = 2, \chi^2 = 0.25, p = 0.88$
- $p > \alpha$ therefore we do not reject the null hypothesis. It appears that the spinner is fair.

7

- The null hypothesis is that the die is fair. $H_0 : p_1 = 0.25, p_2 = 0.5, p_3 = 0.25$
The alternative is that H_0 is rejected and at least one of the following is true:
 $p_1 \neq 0.25, p_2 \neq 0.5, p_3 \neq 0.25$. The significance level $\alpha = 0.05$.

	< 4	4 – 9	> 9
Observed freq	59	180	61
Expected freq	75	150	75

- $df = 2, \chi^2 = 12.03, p = 0.0024$
- $p < \alpha$ therefore we reject the null hypothesis. It appears that the die is not fair.

8

- The null hypothesis is that the claim is correct.
 $H_0 : p_1 = p_2 = p_3 = p_4 = p_5 = \frac{1}{9}, p_6 = p_7 = \frac{2}{9}$
The alternative is that H_0 is rejected and at least one of the following is true:
 $p_1 \neq \frac{1}{9}, p_2 \neq \frac{1}{9}, p_3 \neq \frac{1}{9}, p_4 \neq \frac{1}{9}, p_5 \neq \frac{1}{9}, p_6 \neq \frac{2}{9}, p_7 \neq \frac{2}{9}$
The significance level $\alpha = 0.02$.

	Mon	Tue	Wed	Thurs	Fri	Sat	Sun
Observed freq	25	19	28	22	18	39	45
Expected freq	21.78	21.78	21.78	21.78	21.78	43.56	43.56

- $df = 6, \chi^2 = 3.79, p = 0.71$
- $p > \alpha$ therefore we do not reject the null hypothesis. It appears that the chance of car accidents on the weekend is double that of weekdays.



Critical values

9

- a As $\chi^2_{\text{calculated}} < \chi^2_{\text{critical}}$ we do not reject H_0 .
b As $\chi^2_{\text{calculated}} > \chi^2_{\text{critical}}$ we reject H_0 .

10

- a The null hypothesis is that the ratings do not change.
 $H_0 : p_1 = 0.37, p_2 = 0.25, p_3 = 0.26, p_4 = 0.12$,
 $H_1 : H_0$ is rejected and at least one of the following is true: $p_1 \neq 0.37, p_2 \neq 0.25$,
 $p_3 \neq 0.26, p_4 \neq 0.12$

b

	Poor	Average	Good	Excellent
Observed frequency	26	67	88	110
Expected frequency	107.67	72.75	75.66	34.92

- c $df = 3$
d $\chi^2 = 225.84$
e $\chi^2_{\text{calculated}} > \chi^2_{\text{critical}}$ therefore H_0 is rejected. It appears the improvements have changed the ratings.